The Bifröst

2025 CIV102 Project Team 202: Design Calculations

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Design Calculations for Design 0 under Load Case 1

We have: $V_{max} = 257.33$ N (at $x=0$)

$M_{max} = 69445.33$ N·mm. ($x=644$).

comp on top.

Material Properties

$\sigma_t = 30$ MPa

$\sigma_c = 6$ MPa,

$\tau = 4$ MPa

$E = 4000$ MPa

$\mu = 0.2$

$\tau_{glue} = 2$ MPa

$$\sigma_{max, M, comp} = \frac{My}{I} = \frac{(69445.33)(34.84)}{4.18\times10^5}$$

$$\sigma_{flex, comp, max} = 5.788 \text{ MPa.}$$

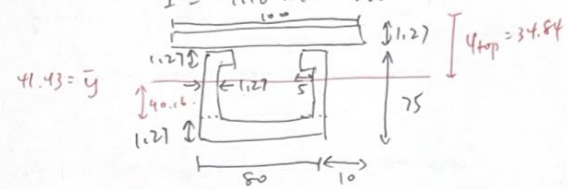
$$\sigma_{flex, tens, max} = \frac{My_{bot}}{I} = 6.883 \text{ MPa}$$

Cross section:

From Deliverable 1:

$\bar{y} = 41.4315$ mm,

$I = 4.18\times10^5$ mm^4.



① Flexural Failure...

$$FOS_{comp} = \frac{6}{5.788} = 1.037$$

$$FOS_{tens} = \frac{30}{6.883} = 4.36.$$

② Shear Failure

$$\tau_{xy} = \frac{VQ}{Ib}$$

Method

$$A = 2\cdot\bar{y}(1.27) + (80 - 2\cdot1.27)\cdot1.27 = 203.6 \text{ mm}^2,$$

$\bar{y}_1 = 0.635$, $\bar{y}_2 = 21.35$, $A_1 = 101.6$ $A_2 = 102.0$.

$$\bar{y} = \frac{1}{A_1+A_2}(A_1y_1 + A_2y_2) = 11.013 \text{ mm},$$

$d = 30.42$ mm,

$Q = Ad = 6193$ mm^3,

$b = 2\cdot1.27 = 2.54$ mm,

$$\tau_{xy} = \frac{(257.33)(6193)}{(4.18\times10^5)(2.54)} = 1.501 \text{ MPa}$$

$$FOS_{shear} = \frac{4}{1.501} = 2.66.$$

Glue. $\tau_{xy} = \frac{VQ}{Ib}$

$A = 100\cdot1.27 = 127$ mm^2,

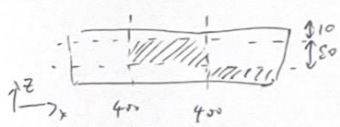
$d = y_{top} - \frac{1.27}{2} = 34.205.$

$b = 10$ mm,

$$\tau_{xy} = \frac{(257.33)(127)(34.205)}{(4.18\times10^5)(10)} = 0.267.$$

$$FOS_{glue} = \frac{2}{0.267} = 7.48.$$

③. Thin Plate Buckling.



Middle section

$$\sigma_{cr} = \frac{4\pi^2E}{12(1-\mu^2)}\cdot\left(\frac{t}{b}\right)^2 = \frac{4\pi^2(4000)}{12(1-0.2^2)}\cdot\left(\frac{1.27}{50}\right)^2 = 3.4545757 \text{ MPa}$$

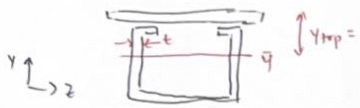
$$FOS_{k=4} = \frac{\sigma_{cr}}{\sigma_{flex, comp, max}} = \frac{3.4545757}{5.788} = 0.597.$$

Side.

$$\sigma_{cr} = \frac{0.425\pi^2E}{12(1-\mu^2)}\left(\frac{t}{b}\right)^2 = \frac{0.425\pi^2(4000)}{12(1-0.2^2)}\left(\frac{1.27}{10}\right)^2 = 23.49 \text{ MPa}$$

$$FOS_{k=0.425} = \frac{23.49}{5.788} = 4.06.$$

③ Thin Plate Buckling Calcs Cont.



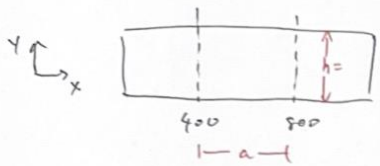
$$y_{top} = 76.27 - 41.43$$
$$= 34.84$$

$$t = 1.27$$

$$\sigma_{cr} = \frac{6\pi^2 E}{12(1-\mu^2)} \cdot \left(\frac{t}{y_{top}}\right)^2 = \frac{6\pi^2 (4\,000)}{12(1-0.2^2)} \cdot \left(\frac{1.27}{34.84}\right)^2 = 27.32 \text{ MPa}$$

$$FoS = \frac{\sigma_{cr}}{\sigma_{flex,\,comp,\,max}} = \frac{27.32}{5.788} = \underline{4.72}$$

Shear Buckling



$$h = 76.27$$
$$a = 400$$
$$t = 1.27$$

$$\tau_{cr} = \frac{5\pi^2 E}{12(1-\mu^2)}\left[\left(\frac{t}{a}\right)^2 + \left(\frac{t}{h}\right)^2\right]$$
$$= \frac{5\pi^2 (4\,000)}{12(1-0.2^2)}\left[\left(\frac{1.27}{400}\right)^2 + \left(\frac{1.27}{76.27}\right)^2\right]$$
$$= (17134.73)(28735 \times 10^{-4})$$
$$= 4.924 \text{ MPa}$$

$$FoS_{shear\ buck} = \frac{\tau_{cr}}{\tau_{max}} = \frac{4.924}{1.501} = \underline{3.28}$$

Summary of FoS, failure mode

$FoS_{flex\ comp}$	1.037
$FoS_{flex\ tens}$	4.36
$FoS_{\tau\ matboard}$	2.66
$FoS_{\tau\ glue}$	7.48
$FoS_{k=4}$	0.597
$FoS_{k=0.425}$	4.06
$FoS_{k=6}$	4.72
$FoS_{shear\ buckling}$	3.28.

∴ It will fail by thin plate buckling in the middle region.

Design Calculations for Final Design under Load Case 2's base case

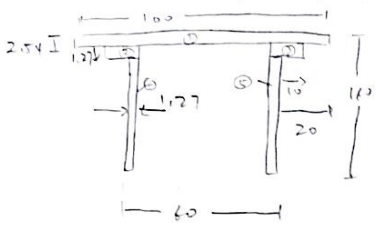
Final Design Calculations for FDS under Load Case 2.

Base Case Weights: 67.5, 67.5, 67.5, 67.5, 91, 91 [N]

By Code, max moment: $M_{max} = -78099.$ N mm (train at 233.7 m)

$V_{max} = 304.1033$ N (train at $x=0$)

Cross sectional Properties of Bridge ($x=30 \rightarrow x=1170$).

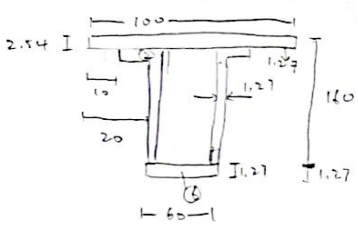


	A_i	$\bar{y}_i$	b	h	I_i	$\lvert y_i - \bar{y}\rvert$
①	254	158.723	100	2.54	136.559	46.55936
②	14.3129	158.095	11.27	1.27	1.924	45.92436
③	14.3129	158.095	11.27	1.27	1.924	45.92436
④	198.3613	79.0475	1.27	156.19	403257.22	33.12314
⑤	198.3613	79.0475	1.27	156.19	403257.22	33.12314

$\bar{y} = 112.17064$ mm

$I = \sum(I_i + A_i(y_i - \bar{y})^2) = 1.8529 \times 10^6$ mm^4

Cross sectional Properties of Bridge ($x=-30 \rightarrow x=30$, $x=1170 \rightarrow x=1230$)



						$\lvert y_i - \bar{y}\rvert$
⑥	76.2	-0.635	60	1.27	10.242	101.42875
①						57.93625
②, ③						57.34125
④, ⑤						21.74625

$\bar{y} = 100.79375$ mm

$I = \sum(I_i + A_i(y_i - \bar{y})^2)$

$= 2.7248 \times 10^6$ mm^4

First, calculate max flexural stress / comp. and shear stress, glue shear stress

Max moment occurs in middle. $y_{top} = 47.82936$,

$y_{bot} = \bar{y} = 112.17064$.

$$\sigma_{flex\ comp.\ max} = \frac{M_{max} y_{top}}{I} = \frac{(78099)(47.82936)}{1.8529 \times 10^6} = 2.01599 \text{ MPa}$$

$$\sigma_{flex\ tens.\ max} = \frac{M_{max} y_{bot}}{I} = \frac{(78099)(112.17064)}{1.8529 \times 10^6 \text{ mm}^4} = 4.72795 \text{ MPa}$$

Max shear occurs at the sides. Assume that max $V(x=30) \approx$ max $V(x=0)$, a conservative estimate. The cs of cross section from $x=-30 \rightarrow x=+30$ is much stronger, so only calculate for the cross section from $x=30 \rightarrow x=1170$.

$$Q = (112.17064)(2 \times 1.27) \cdot \left(\tfrac{1}{2} 112.17064\right) = 15979.46$$

$$b = 2 \times 1.27 = 2.54$$

$$\tau_{max} = \frac{V_{max} Q}{Ib} = \frac{(304.1)(15979.46)}{(1.8529 \times 10^6)(2.54)} = 1.03250 \text{ MPa}$$

Shear stress of glue.

$$Q = (100 \cdot 2.54) \cdot (46.55936) = 11826.07744$$

$$b = 2 \times 10 = 20$$

$$\tau_{glue} = \frac{(304.1)(11826.07744)}{(1.8529 \times 10^6)(20)} = 0.097045 \text{ MPa}$$

Material Properties

$E = 4\,000$ MPa
$\mu = 0.2$
$\sigma_{u,t} = 30$ MPa
$\sigma_{u,c} = 6$ MPa
$\tau_u = 4$ MPa
$\tau_{glue} = 2$ MPa

Bridge Stresses

$\sigma_{flex\ tens,\ max} = 4.72795$ MPa
$\sigma_{flex\ comp,\ max} = 2.01599$ MPa
$\tau_{max} = 1.03250$ MPa
$\tau_{glue} = 0.097045$ MPa

Check Flexural Tension / Comp

$$FOS_{tens} = \frac{\sigma_u}{\sigma_{max}} = \frac{30}{4.72795} = 6.35$$

$$FOS_{comp} = \frac{6}{2.01599} = 2.98$$

Check Shears

$$FOS_{shear} = \frac{\tau_u}{\tau_{max}} = \frac{4}{1.0325} = 3.87$$

$$FOS_{glue} = \frac{2}{0.097..} = 20.6$$

Check Thin Plate Buckling 1 and 2

Refer to Engineering drawings...

$k_1 = 4$, $b_1 = 60$, $t = 2.54$ $\quad \tau_{cr} = \frac{k\pi^2 E}{12(1-\mu^2)} \cdot \left(\frac{t}{b}\right)^2$, $\quad FOS = \frac{\sigma_{cr}}{\sigma_{max}}$

$k_2 = 0.425$, $b_2 = 20$

$$\sigma_{cr,1} = \frac{4\pi^2(4000)}{12(1-0.2^2)} \cdot \left(\frac{2.54}{60}\right)^2 = 24.57.. \qquad FOS = \frac{24.57..}{2.0156} = 12.19$$

$$\sigma_{cr,2} = \frac{0.425\,\pi^2(4000)}{12(1-0.2^2)} \left(\frac{2.54}{20}\right)^2 = 23.49.. \qquad FOS = \frac{23.49...}{2.0156} = 11.65$$

Thin Plate Case 3.

$b_3 = y_{top} = 47.52936$, $\quad k_3 = 6$, $\quad t = 1.27$

$$\sigma_{cr,3} = \frac{6\pi^2(4\,000)}{12(1-0.2^2)} \cdot \left(\frac{1.27}{47.5..}\right)^2 = 14.497, \qquad FOS = \frac{14.497}{2.0156} = 7.19$$

Thin Plate Case 4.

We found diaphragm locations to keep FOS the same, just do edge case.

$k_4 = 5$, $\quad h = 160$, $\quad a = 91$ (overestimate of $91 - 12.5 = 78.5$).

This is what ratio was based out of.

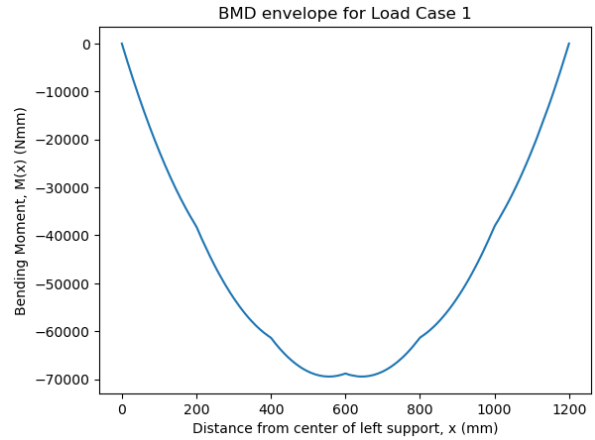
$$\tau_{cr,4} = \frac{k_4\pi^2 E}{12(1-\mu^2)}\left(\left(\frac{t}{h}\right)^2 + \left(\frac{t}{a}\right)^2\right) = \frac{5\pi^2(4000)}{12(1-0.2^2)}\left(\left(\frac{1.27}{160}\right)^2 + \left(\frac{1.27}{91}\right)^2\right) = 4.41691$$

$$FOS = \frac{4.41691}{1.0325} = 4.28$$

Summary Table of FOS:

FOS	Value
flex. tension	6.35
flex. comp.	2.98
mat bond - shear	3.87
glue shear	20.6
thin plate 1	12.19
thin plate 2	11.65
thin plate 3	7.19
thin plate 4	4.28

Code Output for Design 0

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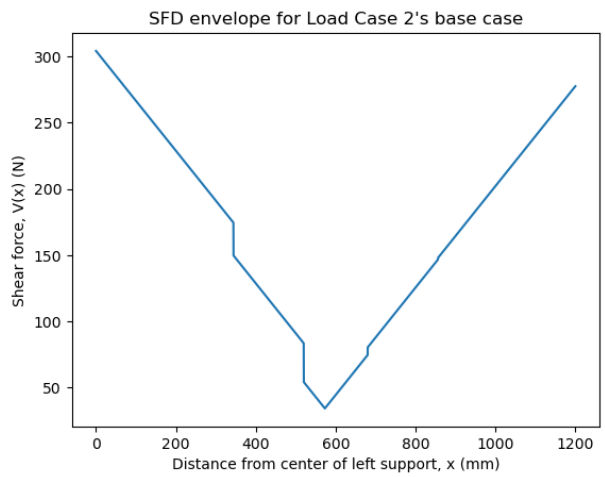
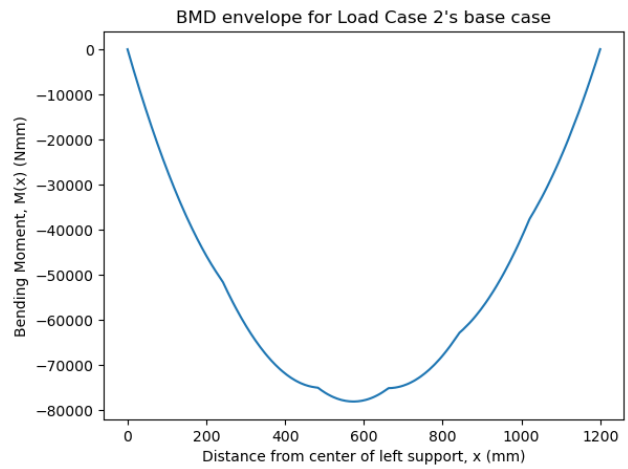
Intermediate Values used in hand calculations
Max Shear:  257.317       at x = 1.000
Max Moment: -69445.333    at x = 128.000
I of cross section:       418000.000
Centroid location:        41.430
Q_max, b_min =            6193.000, 2.540
Q_glue, b_glue =          4344.035, 10.000
```

```
Factors of safety for Design 0 under Load Case 1
FOS, flexual tension:       4.359
FOS, flexual compression:   1.037
FOS, glue shear:            7.479
FOS, matboard shear:        2.665
FOS, thin plate buckling 1: 0.597
FOS, thin plate buckling 2: 4.058
FOS, thin plate buckling 3: 4.720
FOS, shear buckling:        3.280
```

Code Output for Final Design






```
Intermediate Values, for reference
Max Shear:  304.096       at x = 0.020
Max Moment: -78098.993    at x = 233.680
I of cross section:       1878086.550
Centroid location:        111.614
Q_max, b_min =            15821.375, 2.540
Q_glue, b_glue =          11967.361, 20.000
```

```
Factors of safety for final design under Load Case 2's base case
FOS, flexual tension:       6.464
FOS, flexual compression:   2.982
FOS, glue shear:            20.643
FOS, matboard shear:        3.966
FOS, thin plate buckling 1: 12.209
FOS, thin plate buckling 2: 11.675
FOS, thin plate buckling 3: 7.040
FOS, shear buckling:        4.379
```

Description and link to all code files

Link to one-drive file with all code

Calculate.ipynb: This file was used to check final calculations, and provide code output for human-entered parameters

Best.mlx: Contains the iterations and testing for all designs.

Unequalspacing: Calculates for unequal spacing of the diaphragms

Unequallayering: Calculates for unequal thicknesses of the top, sides, and bottom.

Packages installed and languages used: MATLAB, Python, numpy, matplotlib